Distinct Numbers

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1621 | Sorting and Searching | https://cses.fi/problemset/task/1621

You are given a list of n integers, and your task is to calculate the number of *distinct* values in the list.

Input

The first input line has an integer n: the number of values.

The second line has n integers $x_1,x_2,\ldots,x_n$.

Output

Print one integers: the number of distinct values.

Constraints

- $1 \le n \le 2 \cdot 10^5$
- $1 \le x_i \le 10^9$

Example

Input
5 2 3 2 2 3

Output
2